The major, course by course

B.S. MATHEMATICS · PROPOSED: RECOMMENDED BECOMES REQ

- is a prerequisite for
- feeds a course offered only in alternate years
- was "recommended", now required
- offered in alternate years

Hover a course to light up what it needs and what it unlocks; hover an emphasis to see the courses it requires.

WHAT CHANGES

1. Where the bulletin says MATH 23, 51 or 53 is "recommended", it becomes a prerequisite: 23 for 105, 144, 155; 51 for 105.
2. Nothing else moves. No renumbering.

LOWER DIVISION · *all required*

MATH 11
Calculus I
Prereq: MATH 9 / CRE ≥ 76 / AP

MATH 12
Calculus II
Prereq: 11

MATH 13
Calculus III
Prereq: 12

MATH 14
Calculus IV
Prereq: 13

CSCI 10
Intro to Computer Science
or demonstrated proficiency

PHYS 31 · 32
Physics I, II
teaching track may substitute PHYS 11–12

MATH 23
Series & Differential Equations
Prereq: 13 · feeds 105, 141, 144, 155

MATH 53
Linear Algebra
Prereq: 13

MATH 51
Discrete Mathematics
no prerequisite · also CSEN 19

MATH 52
Intro to Abstract Algebra
Prereq: 51

APPLIED · *one of geometry / discrete / applied*

MATH 122
Probability & Statistics I
Pre/co-req: 14 · required in 4 of 5 emphases

MATH 125
Mathematical Finance
Prereq: 53, 122

MATH 141
Mathematical Models
Prereq: 53, 23

MATH 144
Partial Differential Equations
Prereq: 14, 23

MATH 146
Optimization I
Prereq: CSCI 10, 14, 53 · also CSCI 146

MATH 155
Ordinary Differential Equations
Prereq: 53, 23 (CSCI 10 recommended)

MATH 166
Numerical Analysis
Prereq: CSCI 10, 53 · also CSCI 166

MINOR IN MATHEMATICS

MINOR
Mathematics
MATH 11–14, and 52 or 53, plus three 5-unit upper-division courses (not 100, 192, 195)

ANALYSIS · *at least one*

MATH 102
Advanced Calculus
Prereq: 14, 53

MATH 105
Complex Variables
Prereq: 14, 23, 51

MATH 153
Intermediate Analysis I
Prereq: 51 and 102 or 105

MATH 154
Intermediate Analysis II
Prereq: 153 · not on the required list

GEOMETRY · *one of geometry / discrete / ap*

MATH 101
Survey of Geometry
Prereq: 13

MATH 113
Topology
Prereq: 14, 51 (102 recommended)

MATH 174
Differential Geometry
Prereq: 53

ALGEBRA · *at least one*

MATH 103
Advanced Linear Algebra
Prereq: 53

MATH 111
Abstract Algebra I
Prereq: 52, 53

MATH 112
Abstract Algebra II
Prereq: 111 · not on the required list

DISCRETE · *one of geometry / discrete / appl*

MATH 176
Combinatorics
Prereq: 51

MATH 177
Graph Theory
Prereq: 51

DOES NOT COUNT · *toward the seven*

MATH 100 · 190–199
Writing, seminars, research, internship, independent study
MATH 100 is strongly recommended but counts for nothing; 195 satisfies ELSJ only

UNCATEGORIZED · *count toward the seven, satisfy no category*

MATH 123
Probability & Statistics II
Prereq: 122; 53 pre/co · required in 4 emphases

MATH 147
Optimization II
Prereq: 122, 146 · also CSCI 147

MATH 175
Theory of Numbers
Prereq: 52

MATH 178
Cryptography
Prereq: 51 · followed by CSCI 181

MATH 170
Development of Mathematics
Prereq: upper-division standing

MATH 172
Problem Solving
no prerequisite listed

MATH 133 · 134
Logic & Foundations · Set Theory
offered on demand

CSCI 162
Theory of Algorithms
the one CSCI course permitted among the seven

EMPHASES · *optional · at most one · decide by end of junior year*

Applied Mathematics
102, 122, 123 · two of 125 / 141 / 144 / 146 / 147 / 155 / 166 / 178 · one of 166, 147

Data Science
122, 123 · CSCI 10, 60, 61, 62, 183 · CSCI 184 or CSEN 178 · two of CSCI 127/163, MATH 146/147/166, CSEN 166/169, ECON 174

Financial Mathematics
102, 122, 123, 125 · two of 144 / 146 / 147 / 166 · BUSN 70 · ACTG 11, 12 · FNCE 121, 124

Mathematical Economics
102, 122, 123, 146, 147 · ECON 113 · two of 125, 166, ECON 170–174

Mathematics Education
101, 102, 122, 123 (or 8), 170, 175 (or 178), 195 · Urban Education minor advised